

JTrax — Registration form scanning (OCR)

Photograph a paper registration form; the console pre-fills the wizard. Staff confirm before anything saves.

1. Get an API key

Go to aistudio.google.com/apikey, sign in, and click **Create API key**. Copy it once — it is not shown again.

Which tier: the free tier needs no credit card, but Google may use what you send it to improve its products and train models. What you send is a photograph of a child's name, address, date of birth and phone number. **Use a paid key for real registrations** — same code, same endpoint, different terms. The free key is fine for trying it out with a form you made up.

2. Put it in the backend environment

Three variables on **jtrax-backend**. Never commit them; on Render they go in Environment → Add Environment Variable.

```
OCR_PROVIDER = gemini
OCR_API_KEY  = <the key you just copied>
OCR_MODEL    = gemini-2.5-flash          # optional, this is the default
```

Restart the service. With no key set, scanning is simply switched off: the console says so, and the rest of JTrax is unaffected.

3. Use it

Console → **Students** → **Register Student** → **Upload Registration Form** → choose a photo (JPEG, PNG, WebP or HEIC, up to 10 MB). The wizard opens pre-filled.

Fields it fills: name, date of birth, chess level, current school, FIDE ID and rating, and the email and contact number (onto the guardian). Gender, address, how they heard of JCA, previous chess school, course package and the ticked courses are **shown but not saved** — JTrax has nowhere to put them yet.

4. What to expect

- **Nothing is saved by the scan.** It only fills the form; a person still presses Register.
- Anything the model was unsure of is listed under **“Please check”** — look at those before saving.
- A date it cannot read is left blank rather than guessed.
- The photo is not stored: it is read and dropped. The paper form remains the record.
- Only **Admin** and **Reception** can scan.

5. If it does not work

“Form scanning is not configured” — OCR_PROVIDER or OCR_API_KEY is missing, or the service was not restarted.

“Could not read that photo” — usually a real problem with the photo: shoot the whole page straight on, in even light, no shadow across the writing. It can also mean the key is wrong or the free-tier daily quota is spent.

Wrong or missing values — expected on messy handwriting. Type over them; that is what the form is for.

Endpoint: POST /api/v1/registrations/scan · staff only · rate-limited · writes nothing

Swapping provider: implement ocr.Provider in jtrax-backend/internal/ocr and switch OCR_PROVIDER. No calling code changes.